

Integrating CROPWAT Modelling and Deep Learning for Accurate Estimation of Crop Water Requirement

Abstract

Proper estimation of crop water requirements is a significant aspect in management of agricultural water resources, as one of the main strategies that can be used to curb the effects of climatic change on food security. With irrigation taking up a significant portion of the freshwater resources, the need to optimize water-use has become crucial in order to maximize crop yields and minimize water wastage as well as to ensure a sustainable system of agricultural production. The current investigation will seek to estimate crop water demands of wheat crops using CROPWAT 8.0 model and improve the quality of estimation by incorporation of deep learning methods. The climate data was obtained based on the remote sensing data, whereas the soil and crop parameters were obtained based on the Food and Agriculture Organization (FAO) of the selected crops. CROPWAT 8.0 model was used to estimate reference evapotranspiration, crop water requirement, effective rainfall and irrigation water demand. In this study, crop water requirements were computed at 10-day intervals. The optimal irrigation scheduling was determined by identifying critical stages within the soil-moisture depletion (drying) cycle, and the precise irrigation depth required for each irrigation event was quantified. A Convolutional Neural Network (CNN) model was employed to predict the spatial distribution of crop water requirements, resulting in the generation of spatial maps of crop water demand. The optimal CNN model produced overall accuracy 97%, a mean absolute error (MAE) of 0.0052mm/dekad, and a coefficient of determination (R^2) of 0.98. The results indicate that the net crop water requirement for the wheat crop was 219.4 mm during the period from October 2024 to March 2025. The dekadal evapotranspiration rate varied from 23.3 to 60.8 mm/dekad. Furthermore, the CNN model is capable of predicting spatial maps of crop water requirements, while the CROPWAT model is used to compute crop water requirements using agro-climatic parameters. Together, these approaches can be successfully utilized to support efficient water resource management, irrigation infrastructure planning, and the implementation of advanced agricultural technologies, thereby facilitating the effective implementation of government policies aimed at improving water-use efficiency and enhancing crop productivity.

Keywords: Crop Water Requirement (CWR), Convolutional Neural Network (CNN), CROPWAT, Evapotranspiration, Remote sensing, Wheat Crop